

Measuring and Modeling Transient Insulator Response to Charging: the Contribution of Surface Potential Studies

Philippe Molinié

Supélec, Département Energie
Plateau de Moulon
91192 Gif-sur-Yvette Cedex, France

ABSTRACT

What has been the contribution of surface potential studies during the last four decades to our understanding of insulation physics? Which are the promising techniques emerging in this field? The present review covers the industrial context accounting for the development of surface potential measurements on insulating materials, the way they have been implemented, as well as the great variety of models produced to explain the charging and potential decay mechanisms. Several polarization or transport processes can be responsible for the decay of the potential. Though most of the models initially stemmed from electrostatics and semiconductor physics, around the notion of mobility, experiments on polymers required “thermodynamic” models, describing progressive charge detrapping. We also underline here that the three main different physical processes likely to be involved in the potential decay (dipolar relaxation, dispersive transport, slow detrapping) can lead in disordered materials to the same time response, the challenge being here is to design inventive new procedures to distinguish them. This comment also applies to most of electrical measurements on disordered materials. Kelvin probe microscopy, return voltage and thermostimulated potential measurements also illustrate the multiform development of surface potential measurement techniques, as their future prospects.

Index Terms — Charge injection, charge measurements, dielectric polarization, modeling, nondestructive testing, polyethylene insulation.

1 INTRODUCTION

Understanding the transient response of an insulator submitted to the swift application of a dc field is of practical great importance, since this transient response can last a very long time before a steady regime is reached, especially concerning polymers, where slow relaxation processes, slow charge movement and slow charge detrapping usually dominate the response. That has to be kept in mind when performing electrical measurements.

Charge mapping techniques, either thermal or acoustic, have brought during the last twenty years a huge amount of data concerning charge distribution and transport in the insulators.

A comprehensive review of these data had been written by Sessler several years ago [1]; since then, new focused techniques (FLIMM, for instance) made possible 3D charge density mapping [2].

Analyzing these experimental data in terms of homo- and heterocharge, it is in principle possible to discriminate charge injection from dipolar polarization. However, the observed charge patterns are the result of transient processes occurring during the first stages of poling, which are often difficult to visualize, since a minimum delay is necessary, using these techniques, to record a usable charge map. Next, charge carriers recognition remains a difficult question.

For these reasons, cross-checking the results with other techniques, either isothermal or thermally stimulated, is necessary.

Manuscript received on 1 February 2005, in final form 6 July 2005.

To have an insight into fast transient phenomena, current measurement during the poling stage remains the simplest technique. Surface potential measurements can also be used: when coupled with constant current charging devices, build-up of the potential can be monitored, so that recording part of the fast phenomena occurring during the poling stage is possible [3]. Compared to current measurements, surface potential techniques have the advantage to give, for slow transient phenomena, a stable signal, since they are sensitive to charge position, whereas current measurements are recording charge displacement. Besides, they can be used for films of thickness below micrometer [4], where most of the present charge mapping techniques fails.

We present here mainly a synthesis of surface potential decay studies, which relies on the analysis of a bibliography of nearly 200 papers, built up from an initial fund gathered by R. Coelho along the 70s and the 80s, which we have enriched since then. The present work cannot however pretend to exhaustiveness, as the electrostatic probe worldwide diffusion since the sixties and seventies has led to numerous works throughout the world. We limited our study to papers presenting sizeable modeling, or experimental results which notably influenced model development.

1.1 POTENTIAL PROBES

1.1.1 CAPACITIVE TYPE PROBE

Simple in its principle, the capacitive probe is made up by a conductive plate on which an influence charge is induced by the studied surface. Influence depends on the probe distance to the surface; besides, probe voltage remains in principle close to zero, so that it produces on the studied surface a field depending on the probe distance, which might perturb the potential decay process in a non-reproducible way.

This principle however allowed the development of a lateral high resolution (about 80 micrometers) probe, which has been used by researchers in Bangor in the 1970s [5-7], and led to good results for potential mapping. One can also mention the “field mill”, which allows, through the rotary motion of a star-shaped electrode behind a grounded electrode cut out with the same shape, to measure the absolute field, which avoids the inevitable drift of the capacitive probe. But in both cases, what is performed is rather field measurement, than potential measurement.

1.1.2 KELVIN TYPE PROBE

The most common electrostatic probe model is based on the vibrating reed principle, with a feedback loop that allows setting the probe at the surface potential. The major advantage of the system is to perform voltage absolute measurement, with a minimum perturbation of the surface.

Decay models usually consider the electric field to be zero above the surface, which is effectively achieved using this kind of probe.

The development of atomic force microscopy led, a few years ago, to microscopic Kelvin probes, that can be used to obtain high resolution (up to 10 nm) maps of the surface potential distribution on non-conductive materials [8, 9]. In Kelvin probe force microscopy (KFM), the tip movement is analyzed by two feedback loops, the first to control the distance from the probe to the surface, and the second to minimize the electric field between them.

1.1.3 ELECTRON BEAM PROBES

After charging the insulator by electron implantation in a scanning electron beam microscope, potential measurements can be performed by measuring the deflection of a low energy electron beam parallel to its surface [10]. The “electrostatic mirror method” can also be used for computing charge decay from the electrostatic mirror contraction, as described in [11].

1.2 INDUSTRIAL CONTEXT

Three main fields of industry led to most of the studies involved in our review.

1.2.1 ELECTROOPTICS

Since 1950, the design of photocopiers and laser printers led to much of groundwork, issued by the big American industrial research laboratories, as IBM [12], Xerox [10, 13], or Kodak [14], on corona charging, charge injection mechanisms, and on potential decay dynamics, especially on photoconductive insulators. Taking Rose papers, and semiconductor physics tools, as a starting point, the decrease of the potential due to transport mechanisms in a disordered material has been modeled in detail, by introducing trapping, which led to the dispersive transport models and to the thermodynamic models that we present further on.

1.2.2 ELECTRETS

The feasibility to store charges for a long time in insulating materials has been exploited to develop many electret-based devices, especially in the field of electroacoustics. Charge decay mechanisms were studied here also, but from an opposite point of view, which was to improve the time stability of the charge. Therefore these teams, among them researchers from Darmstadt [15-17] and Sao Paulo [3, 18-20], concentrated on the topic of charge build-up and trapping into the material, rather than on decay models, even if they published a few of them (see for instance [19], and [15] for PTFE, or [16] for SiO₂). They designed original charging devices, which we will mention later on.

1.2.3 ELECTRICAL ENGINEERING

1.2.3.1 CHARGE INJECTION STUDIES, FOR CABLES AND HIGH VOLTAGE INSULATION DEVELOPMENT

The multiform development of insulating polymers for high voltage insulation has also been the source of numerous works on surface potential, in Japan (especially in Nagoya [21-23]), France (at Supélec, for instance [24-29]) or Canada [30-34].

We will only mention here one result which had an important influence on potential decay models: the observation published by Ieda in 1967 [21], that, on polyethylene, in particular circumstances, potential decay curves obtained on the same material with several initial charge levels did cross each other. This cross-over phenomenon has been the scope of many works, and excited theoreticians imagination [20, 24, 30, 32, 35, 36]. The most convincing explanation of the phenomenon has in the end been published by Perlman and his co-workers from Quebec [32].

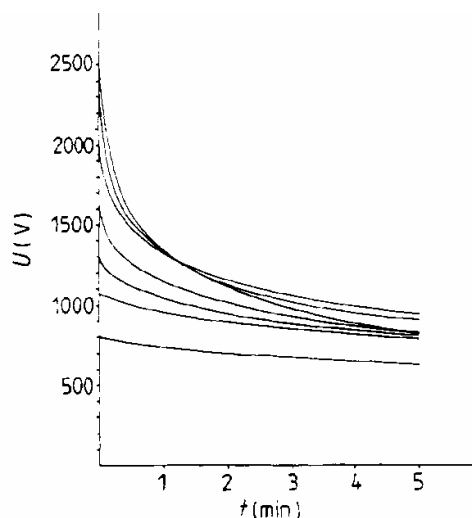


Figure 1. Potential decay curves cross-over [36].

1.2.3.2 RETURN VOLTAGE TECHNIQUES FOR AGING MONITORING IN POWER TRANSFORMERS AND CABLES

The value of the return (or recovery) voltage after dc charging and temporary short circuit on impregnated dielectrics is very sensitive to the moisture content of the insulator, and provides a good tool for aging monitoring in transformers (see [37] and Figure 2), as well as in cables [38].

This industrial need led to interesting modeling of potential decay and return due to the very low frequency

component of their dielectric permittivity [39]. This model usually assumes the insulator to be linear and described by a response function; from this point of view, potential decay and return studies are part of time domain dielectric spectroscopy, together with absorption currents.

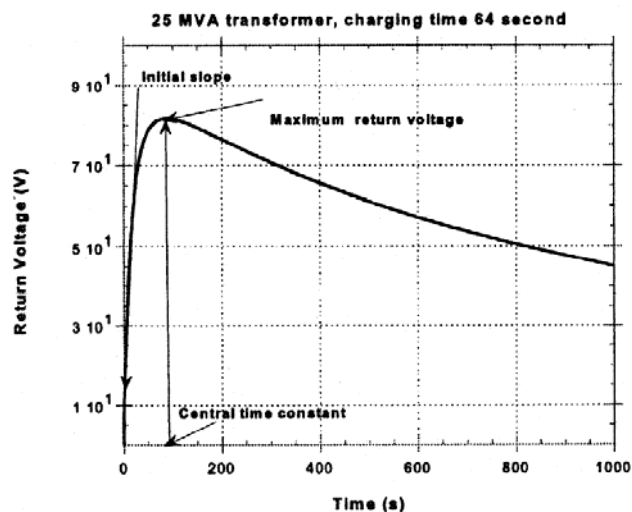


Figure 2. Typical return voltage on a retired transformer [37].

1.3 INSULATOR CHARGING MECHANISMS

The main charging processes of an insulator are by contact (either by a liquid or by a solid), corona discharge, and electron beam. In the latter case, charge is directly injected, with a defined energy, into the bulk of the material, and its drift is depending on the trapping properties of the material. Contact deposited charge, on the opposite, is transferred on material electronic surface states. As for corona charging, charge is deposited as positive or negative ions, and will equally be transferred, through mechanisms that have been studied in details on selenium by Vance [13]; this electron or hole transfer into the insulator electronic surface states may involve Auger mechanisms in positive (accompanied by additional electron-hole pair creation). However, should charge be transferred by contact or corona, surface states usually are deep traps in the forbidden energy band of the insulator, so that an additional energy will be needed to inject charge into the volume electronic states.

From experiments, it has actually been known for a long time that potential decay may depend on the charging process, and not only on the amount of deposited charge, which reveals the role of the energy factor in charge injection phenomena. A 1968 paper from Ieda [22] was already showing, for instance, that the deposit of a conductive paint on a polyethylene film was slowing down the decay after corona charging. The injecting effect of the corona discharge was however really proven by a team

from Bangor in 1977 [5], and in the end described in detail by Perlman and his team from Quebec, who showed that the neutral particles of the negative corona discharge [33] and especially activated nitrogen oxides [34], are mainly responsible for the injection.

Let us remember also that, even without this corona discharge energy activation, the injection of a trapped electron as a molecular ion cannot be ruled out. It is even a very likely phenomenon in polymers.

Experimental measurements of charge build-up have been performed in many materials by researchers working on the field of electrets [17, 20], using in particular constant current corona triodes, developed in the seventies by Moreno and Gross [18, 3].

2 POTENTIAL DECAY MODELS

2.1 INTRODUCTION

Let us consider an insulator as a large plate of thickness d small compared to its other dimensions, so that side effects can be neglected. A one dimensional model of the potential decay can then be proposed, where the charge densities, field and potential are function of the distance to the ground z only. The boundary conditions are that the lower surface is grounded, and it usually is assumed that the electric field is zero at the upper surface (which is indeed established in the Kelvin case). Considering for instance the electric displacement $D = \epsilon_0 E + P$ (P being the polarization inside the insulator), its value in the insulator close to the upper surface is then equal to the free charge density $q(t)$ deposited on this surface.

Modeling the potential decay experiment, the insulator is usually assumed to be charged instantaneously at $t=0$, and potential monitored immediately afterwards. During the potential decay, we have an open circuit, so that the continuity equation can be written in any point of the insulator, as:

$$\frac{\partial D}{\partial t} + \left(\sigma + \sum_i \mu_i \rho_i \right) E = 0 \quad (1)$$

The first term is the time derivative of the electric displacement. The second term is related to the electric conduction, σ being the intrinsic conductivity, μ_i and ρ_i being the mobility and charge density of a particular injected charge carrier, E being the electrical field. Diffusion is in this case assumed to be zero.

We will examine separately in next sections the influence of these three physical mechanisms on the decay.

2.2 DIPOLAR POLARIZATION

Dielectric relaxation of most insulating materials, among them almost all the polymers, reveals a very low frequency component, linked as well to internal molecular reorganization (polymer chains slow movement) and to complex interfacial polarization phenomena. This response can be modeled in the linear regime by a dielectric response function $\phi_D(t)$, establishing a convolution relationship between the electrical displacement and the field [40] :

$$D(t) = \epsilon_0 \int_{-\infty}^t E(\tau) \phi_D(t - \tau) d\tau \quad (2)$$

or by a function $\phi_E(t)$ linking electric field and displacement :

$$E(t) = \frac{1}{\epsilon_0} \int_{-\infty}^t D(\tau) \phi_E(t - \tau) d\tau \quad (3)$$

ϕ_D and ϕ_E are non independent, the product of their Laplace transforms being 1.

In the case of an homogeneous dielectric, with no space charge, the electrical field is constant in the insulator, and

can be related to the potential by $E = \frac{V}{d}$.

Equations (2) and (3) can then be transformed to link measurable values:

$$q(t) = \frac{\epsilon_0}{d} \int_{-\infty}^t V(\tau) \phi_D(t - \tau) d\tau \quad (4)$$

$$V(t) = \frac{d}{\epsilon_0} \int_{-\infty}^t q(\tau) \phi_E(t - \tau) d\tau \quad (5)$$

Assuming then the conductivity to be zero, the continuity equation (1) implies that, during the decay, $\frac{\partial D}{\partial t} = 0$, so that

the free charge density on the surface is a constant. It can be written $q(t) = \Gamma_0(t) q_a$, $\Gamma_0(t)$ being the Heavyside function centered on $t=0$.

From equation (5) we can deduce:

$$\frac{dV(t)}{dt} = \frac{dq_0}{\epsilon_0} \frac{d}{dt} \int \phi_E(\theta) d\theta = \frac{dq_0}{\epsilon_0} \phi_E(t) \quad (6)$$

This relationship implies that the material does not react instantaneously to the changes of the electric field. If a charge is deposited on the surface at a given time, a potential step will follow, linked to Poisson relationship, screened by instantaneous polarization processes, and then, potential will

decrease under the influence of slow polarization, this phenomenon being numerically computable through inversion of the equation. The general theory of this phenomenon, in the case of both potential decay and return, was established by Gross in 1942 [41], and a recent model has been given in [39].

Some remarks have to be done:

a) deducing $\phi_E(t)$ function from dV/dt is easy, since, assuming $V_0 = \frac{dq_0}{\epsilon_0}$, then $\phi_E(t) = \frac{dV(t)}{V_0 dt}$ (for $t \neq 0$, $\phi_E(t)$ is negative).

However, we have to underline that V_0 is higher than the charging potential (potential at the end of the charging process), or than the first measured value of the potential, which can be calculated as $V_t = V_0 \epsilon_0 / \epsilon_t$, (ϵ_t being the permittivity value taking into account all the instantaneous and rapid polarization processes having taken place before t – i.e. at the end of charge, in the former case, or when the first measurement is performed, in the latter). Because of the delay necessary to charge the sample, V_0 is also higher than the charging potential. Hence it is better to have a direct measurement of the deposited charge q_0 .

b) the response functions that can be found in the literature are usually given for the constant field case, corresponding to absorption currents measurements. In these, a potential step is assumed to be applied at $t=0$ so that: $V(t) = \Gamma_0(t)V_0$. From this experiment, can be easily deduced the value of $\phi_D(t)$ since (assuming again the absence of any conductivity):

$$i(t) = S \frac{dq}{dt} = \frac{\epsilon_0 S V_0}{d} \frac{d}{dt} \int \phi_D(\theta) d\theta = C_0 V_0 \phi_D(t) \quad (7)$$

(S being the insulator surface, and C_0 the geometrical capacitance of the insulator)

The dielectric response function $\phi_D(t)$ is then proportional to the absorption current.

Besides, for reasons studied in particular by Jonscher [42], the usual response function in condensed matter is composed of time power laws, as is shown in Figure 3. (The usual $f(t)$ notation in dielectric relaxation studies refers to the convolution relationship between electrical field and polarization, whereas $\phi_D(t)$ that we have been chosen is involved in the relationship between displacement and polarization. The two functions are linked by: $\phi_D(t) = f(t) + \delta(t)$, δ being the Dirac function, so that the shape given for $f(t)$ in Figure 3 would be identical for ϕ_D).

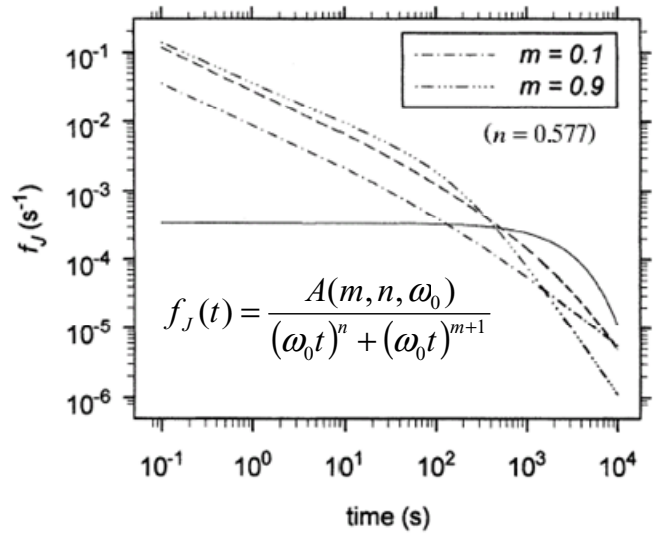


Figure 3. Jonscher response functions [43].

In this case, potential derivative, proportional to $\phi_E(t)$, will also be composed of time power laws, with different coefficients [26, 39, 43], and will bear the same aspect using $\log(dV/dt) = f(\log t)$ plot than the response functions shown on Figure 3, so that it also will be represented by one or two line segments, with slopes more or less close to unity. The exact treatment to derive $\phi_E(t)$ from $\phi_D(t)$ involves an inverse Laplace transform calculation.

We have been publishing examples of this kind of behavior in epoxy [26], polypropylene [27-29], and polyethylene [27]. Das Gupta [7] also attributed decay on polyethylene films to slow polarization phenomena.

2.3 INTRINSIC CONDUCTION

2.3.1 INTRODUCTION

Assuming now that polarization is stabilized, so that the material can be modeled by its constant permittivity ϵ , and neglecting again space charge effects, the existence of an intrinsic conductivity σ transforms equation (1) into $\sigma E + \epsilon \frac{\partial E}{\partial t} = 0$, so that :

$$\frac{dV}{V dt} = -\frac{\sigma}{\epsilon} \quad (8)$$

A constant intrinsic conductivity will then lead to an exponential shape of the potential decay, with a time constant σ/ϵ . The conductivity being proportional to the product of the mobile charge carrier density by their mobility, potential decay models assuming intrinsic conductivity rely on particular hypotheses on carrier generation, recombination, and mobility. We will only describe here a few of these models.

2.3.2 HYPOTHESES ON CARRIER GENERATION

Generation of the charge carriers can be due to injection from the electrodes: this case will be examined in section 2.4. Otherwise, electron-hole pairs can be created through band ionization, but, in insulating materials with gaps greater than 5 eV, this can only be performed by irradiation. A model of potential decay due to gamma ray-induced conductivity was for instance proposed by Frederickson to account for experimental results in polyimides [44]. A model of the decay following photoexcitation of the carriers has also been developed by Batra for xerography applications [12]. However, the latter is closer to a charge injection model than to an intrinsic conductivity model, since it assumes a short distance penetration of light into the insulator, so that electron-hole pairs are generated close to the charged surface only, the surface-deposited charge being then neutralized by internal charge of the opposite polarity, while the remaining internal charge of the same polarity will drift into the material.

Charge detrapping from shallow traps can also be assisted by temperature and field through Poole-Frenkel effect at high fields. That will lead to a field-dependent conductivity, and potential decay has been modeled according to this mechanism [45]. An analog process is ion pairs creation through field or temperature-assisted dissociation of impurities.

2.3.3 HYPOTHESES ON RECOMBINATION-EXHAUSTION

Due to retrapping, the free path of electrons and holes is usually very short in the disordered materials used in insulation, so that the intrinsic conductivity is always very low, even when considering the generation mechanisms listed above. Furthermore, a particular mechanism has been assumed by Malecki [46] to explain the great stability of Teflon FEP electrets: field-assisted carrier exhaustion. This phenomenon should be observed assuming the mean time of flight to be less than the conductivity recovery time due to charge generation, and will lead to a current density determined by the generation rate of the carriers only, so that the decay rate will increase linearly with the thickness of the sample, and will tend towards a constant for long time periods (months or years).

2.3.4 SURFACE CONDUCTION

Due to their very low volume conductivity, surface conduction often predominates on polymers. Thus, several potential decay models on insulators take into account this parameter. They were developed for surface aged insulators in presence of humidity [47]. However, in this case, a truly conductive surface layer is involved, which properties are very different from the material itself and which can produce

an exponential potential decay. Such a behavior is indeed very often the sign of surface conduction. However, the reverse is not true: cases are reported of potential mapping experiments showing charge lateral spreading [6], but without leading to an exponential decay.

2.4 CHARGE INJECTION AND TRANSPORT

Most of the developed models are based on the hypothesis of surface charge injection into the bulk, and on various physical hypotheses concerning transport.

2.4.1 ELECTROSTATIC MODELS

We classify under this designation transport models which basically do not take into account the influence of charge trapping and detrapping processes on the charge decay mechanism, considering therefore transport only by means of a carrier mobility μ . Within this theoretical frame, from the boundary conditions, various models can be established. Detailed calculation has been established by Batra et al. [12], who computed in the general case decay equations. The main idea, in the “Kelvin” case of a zero field outside the insulator, is that charge distribution may be considered as successive sheets, parallel to the surface, which are driven by a constant field along their drift through the material, so that each of them moves forward at constant speed (even if mobility is field dependent). If instantaneous charge injection from surface to bulk electronic states is assumed to occur during the charging phase, these models therefore predict a linear potential decrease as long as the charge front has not reached the grounded counter electrode; they also imply that the initial potential decay rate is proportional to the square of the initial voltage. After the charge front transit time, the voltage versus time plot presents a bend, the number of charged sheets contributing to the decay decreasing gradually. The last charged sheet is driven by a zero field, and hence moves at zero speed, so that it will stay on the upper surface.

This kind of model was also developed at that time by Wintle [30], who included diffusion and conduction, and later by Coelho [24, 25, 35], Wintle model being then developed by Hill [48] (a review by Coelho can be found in [35]). However, despite the interest of these models for semiconductors, their application field seems to be quite limited concerning insulating materials, since we were not able to find any paper reporting in an indisputable way, on real insulators, a behavior corresponding to these calculations. What is usually found is shown on the upper part of Figure 5, so that the linear initial decrease that the models predict, which would lead to an horizontal straight line on the plot, is never observed. Let us remark that taking into account a field depending mobility [14, 35] does not change this prediction of the model.

Assuming a steady-state equilibrium to be achieved, and considering the mean time τ_t spent by the charge in trapping levels, compared to the duration τ_0 of its stay in transport levels, it is also possible to define an effective mobility taking into account trapping processes, which will be $\mu_{eff} = \mu \tau_0 / (\tau_t + \tau_0)$ [49]. Nevertheless, it obviously does not change anything to the shape of the $V=f(t)$ curve, produced by the model.

2.4.2 ELECTROSTATIC MODELS INCLUDING SURFACE RETENTION AND FIELD DEPENDENT EMISSION

After some controversy, it became clear in the end of the seventies that the electrostatic models presented above could not account for the cross-over phenomenon, whose only coherent explanation, given by Sonnonstine et Perlman [32], was partial surface charge retention, surface emission into the bulk being determined by an injection law depending on the field. According to Sonnonstine's calculations, two successive regimes occur within this theoretical frame : an high field regime, limited by the drift of the internal space charge, and a low field regime, limited by the surface charge retention, the cross-over being due to the existence of the first regime only.

This kind of modeling introduces in potential decay models the notion of surface energy barrier, and field-assisted injection over it. We already mentioned this point in section 1.3, and we will discuss it in details in section 2.5.

2.4.3 MODELS COMBINING ELECTROSTATICS AND THERMO-DYNAMICS. INFLUENCE OF TRAPPING AND DISPERSIVE TRANSPORT ON VOLTAGE DECAY DYNAMICS

A correct treatment of the potential decay problem in a disordered material actually requires that trapping during charge drift should be taken into account, since it will greatly influence the potential decay shape. A first line of reasoning, modeled by Wintle [31] (and taken up later by [19]), considers an effective mobility for carriers, though restricting it to hopping through transport levels between shallow traps only, while irreversible trapping in deep traps is introduced. This kind of models predicts a decay tending towards a constant, non-zero value, of the potential, and is closer to experimental results in insulators than the models reviewed above.

Later on, Von Berlepsch [36] produced a more complete treatment for this problem by introducing reemission of charge by deep trapping levels, together with field-assisted surface charge emission according to Sonnonstine's model. He also defined the concept of a time dependent effective mobility. Recently, experimental data on polypropylene were confronted with a model and computations along the same lines [50].

However, it is not possible nowadays to treat the question of charge transport in a disordered medium, without mentioning the more general frame of dispersive transport, which handles the question as a whole, by assuming a continuous energy distribution of the trapping levels. Since Scher and Montroll paper [51], many transport models in disordered systems have been proposed. It is possible to consider numerical computing of this multiple trapping problem (an example can be found in [10]).

A simplified approach introduces the concept of demarcation energy, defined as:

$$E_d(t) = kT \ln[(N_c/N_t)(t/\tau_0)] \quad (9)$$

(N_c being the conduction states density, N_t the trap density, and τ_0 the carrier lifetime in conduction states).

The hypothesis is that, at a particular time t , the emission of traps whose energy is below E_d can be neglected, whereas one can consider that higher energy states are in equilibrium with transport states. An analytical computation of the potential decay along these lines was performed by Arkhipov [52] (see also [53]). Assuming an exponential shape of the trap distribution, he showed that a $\log(dV/dt)=f(\log t)$ plot was tending towards two straight lines with different slopes, on either sides of a mean transit time value (Figure 4).

However, one has to remember that this approximation, assuming equilibrium between shallow states and transport states, relies on the fact that the lifetime of carriers in transport states is negligible compared to the dwell time at the demarcation energy, so that the model is not valid at short times, when this equilibrium is not reached.

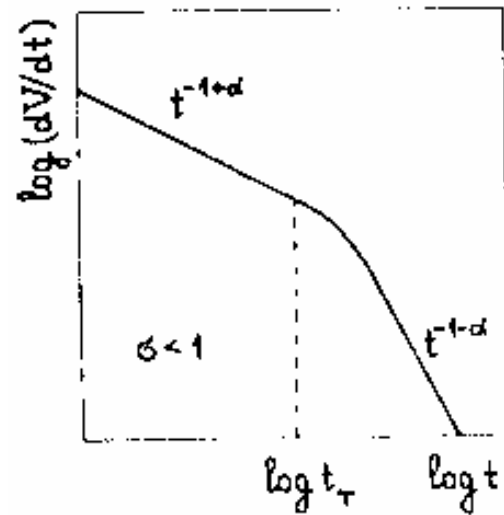


Figure 4. Potential decay due to dispersive transport [51].

2.4.4 PURELY THERMODYNAMIC MODELS : DETRAPPING CONTROLLED DECAY

In the case where the carriers transit time is low compared to their mean characteristic detrapping time, the geometrical factor can be completely eliminated from the models, detrapping kinetics being then the main factor determining the shape of the decay, rather than charge displacement. Trapping charge stability is proportional to the exponential of the trap energy depth, so that the slowdown observed in the potential decay rate can be attributed to the progressive increase in the trapping energy depth of the carriers. The demarcation energy considered above can thus be considered as the limit between filled deep traps and shallow traps, emptied since the beginning of the potential decay. Using this approximation, formulated by Simmons and Tam [54], and applied by Watson [10] to potential decay modeling, detrapping emission current is linked to the trap energy density $N(E_d)$ at the level $E_d(t)$ in the following way :

$$I(t) \propto N(E_d(t)) \frac{dE_d}{dt} = N(E_d(t)) \frac{kT}{t} (N_t/N_c) \tau_0 \quad (10)$$

One can deduce from this equation that the plot of time multiplied by intensity $tI(t)$, versus the time t , will give using a semilogarithmic scale an image of $N(E)$, the trap density versus their energy depth. Concerning the potential decay experiment, the decay rate dV/dt will be proportional to the right part of equation (10), so that, multiplied by time ($t dV/dt$), it will give the same image of $N(E)$. Watson has shown that this simple plot, compared to a detailed numerical calculation, underestimated the shallow traps density compared to deep traps, since it neglects the influence of retrapping during the charge drift. However, it can give interesting results, which he presented on monodisperse polystyrene. On Figure 5, we reproduced from [10] the potential decay rate recorded on this material, together with $t dV/dt = f(\log t)$ plot, whose abscissa has been converted in energy through comparison of curves at various temperatures.

2.5 SURFACE DETRAPPING LIMITED DECAY

As has been shown by Sonnonstine model (see section 2.3.2), the hypothesis of a field-dependent injection leads to a transition from a volume-limited regime at high fields to a surface-limited regime at low fields. This is in good agreement with thermostimulated currents measurements which, according to ideas initially expressed by Von Seggern [55], may allow to discriminate surface from volume traps and have shown that, on most of insulating materials, surface traps were deeper in energy than volume traps.

Therefore, situations where surface progressive detrapping is determining the dynamics of the decay on the long time

should be numerous, though this hypothesis is practically never suggested in the literature. In a recently published article ([29], see also [28]), we have shown that a model of progressive surface detrapping was the only way to account for measurements performed in Supélec on 6 μm polypropylene films, negatively charged by corona, at high fields and ambient temperature.

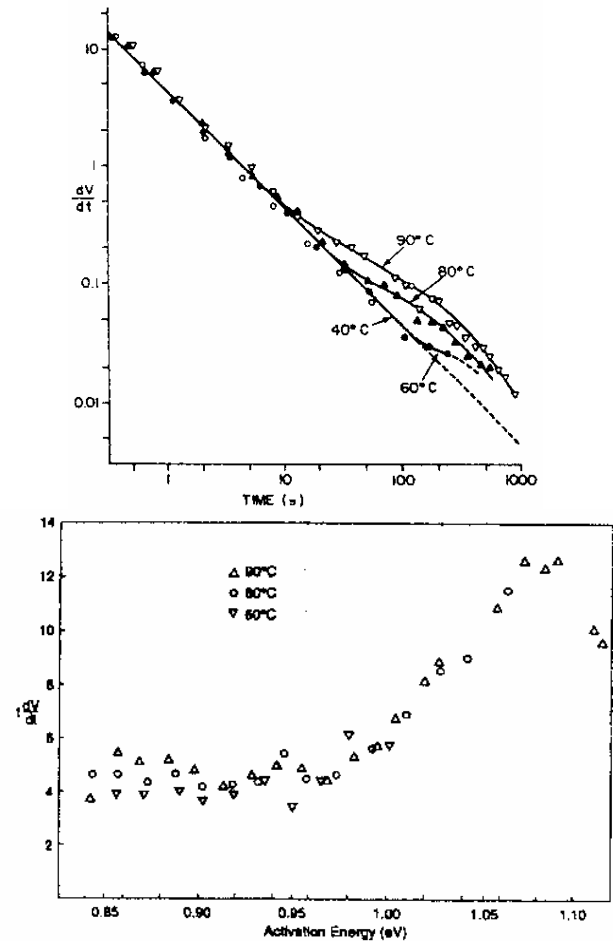


Figure 5. Data treatment according to the demarcation energy model [10].

The transition from a volume-limited regime to a surface-limited regime may explain some of the published data on potential decay. For instance, Von Berlepsch [36] ascribes to the predominant influence of surface trapping the fact that 15 μm high density polyethylene films present a slower decay than 20 or 30 μm films, mentioning Von Seggern results showing that the surface region can extend to a 5 μm depth [55]. This explication only is however not satisfactory, since the amount of surface traps will be identical in both cases, so that surface detrapping only cannot explain the difference observed. However, assuming that, for the thickest films, the first part of the decay is volume-limited (i.e. dominated by the relatively quick charge drift by hopping between volume

traps), while the decay is entirely dominated by surface detrapping on the thinnest (probably after a volume-limited phase occurring during the charging period) provides a possible explanation for a slower decay on thin films. This would also be coherent with our measurements in Supélec.

2.6 WHICH MODEL FOR WHICH MATERIAL?

Having reviewed the various existing models, we are coming to this question, which is most important from a practical point of view. However, it is a tricky one, since we have shown that three different classes of physical phenomena can produce the same power-law time dependence for the decay rate, leading to straight lines on the $\log(dV/dt)=f(\log t)$ plot, with a mean slope close to -1 , but with a bend making possible to distinguish two domains, with different slopes. This bend could be interpreted as the center of a dipolar relaxation time distribution, in the frame of a model where deposited charge does not inject (Figure 3), or as a transit time in a transport model (Figure 4), or as a peak of a trap energy density distribution in a thermodynamical model (Figure 5).

Practically speaking, we suggested elsewhere [28,29] to systematically choose representations giving prominence to the decay dynamics, and we proposed the use of the tdV/dt versus $\log t$ plot, which has been developed within the frame of demarcation energy models, as being the most appropriate for the polymers used in insulation or in electrets, independently of any hypothesis on the physical mechanism involved (for instance a bend or a peak can be a transit time, as in Figure 4, or a particular characteristic relaxation time as in Figure 3). Using $V=f(t)$ plot, it is much too easy to perform a very good fit of a model curve with the experimental one ; hence that kind of representation is not satisfactory.

Next, to check the possible existence of a conduction in the material, $\log(dV/Vdt)$ versus $\log t$ plot has to be employed, since, using this representation, constant intrinsic conduction should lead to an horizontal asymptote. Here again, let us remark that the choice of the logarithmic scale on both axes is clearly necessary, otherwise, in a linear plot, the false feeling of a current stabilization will be induced.

Last, multiple experiments have to be performed, involving changes of the interfaces, of the thickness, changing the voltage and polarity, together with comparison with charge mapping techniques, to check what kind of model (polarization, transport, detrapping, or surface conduction) is appropriate [56]. Potential scans can help to evaluate the importance of charge lateral spreading, or surface conductivity, and return voltage measurements can also provide a lot of information [26].

Our experience in the field of potential decay experiments on insulating materials is that polarization phenomena are too often neglected by authors, and, concerning charge injection, literature shows a tendency to forget the hypothesis of ion injection, and to overestimate electronic transport phenomena. Both tendencies being most likely linked to the historical preeminence of models coming from semiconductor physics ; we suggest that one has to be careful before applying them to polymers, where electronic effective mobility is very low, and where dipolar and ion phenomena will hence often predominate.

3 TECHNIQUE DEVELOPMENTS

Surface potential measurements technique has been developed in several ways. However, having chosen to focus this review on the topic of isothermal potential decay modeling on insulators, we will limit to return voltage, and thermostimulated surface potential, which both can easily be performed together with potential decay experiments, and provide valuable information to check the validity of decay models.

3.1 RETURN VOLTAGE MEASUREMENTS

The existence of a return (or recovery) voltage on capacitors charged during a long time by a dc voltage, and then rapidly short-circuited, has been known for a long time. This return voltage can give information on conduction and polarization mechanisms on the insulators and is used nowadays as an aging monitoring tool (see section 1.2.3.2). Among the phenomena described above in section 2, only slow dipolar polarization and charge injection can lead to return voltage.

The theory of return voltage due to dipolar relaxation has been first described by Gross [41], and more can be found in recent articles [39, 43, 26]. Models of return voltage due to space charge have been developed by Coelho [25], by considering an injection as sheets of charge deriving parallel to the surface. We intend to present a more detailed treatment in a future article, along the following ideas: when an insulating sample with a given internal space charge distribution is neutralized, the average field in the sample becomes zero, so that at least a zero field plane will necessarily appear in the bulk (one in the case of unipolar injection, two if injection is bipolar). The fraction of the injected charge having passed beyond this plane will continue its injection drift, whereas the rest will move back to the injecting electrode. If most of the charge is located near the injecting electrode, which is often the case, then most of the charge will be located on the "return side" of the zero field plane, and will return to the injecting electrode, so

that there will be a return voltage of the same sign than the applied voltage.

3.2 THERMOSTIMULATED SURFACE POTENTIAL

Thermostimulated surface potential (TSSP), or thermostimulated charge decay (TSCD) measurements are closely related to thermostimulated currents [57]. For TSSP as for TSC, the insulator is poled at high temperature, then cooled without removing the voltage, then gradually heated, the potential decay being measured instead of the current. The results of both experiments are linked by:

$$j_{TSC}(T) = -C\beta \frac{dV_{TSSP}(T)}{dT} \quad (11)$$

β being the heating rate, and C the geometric capacitance of the system

TSSP has been used for polymer characterization since the early eighties in Japan, especially in Nagoya (see for instance [23]) and has also been used for studies on electrets stability [58] or piezoelectricity [59]. Recently, it was proven useful for thin (less than 1 micrometer) dye doped polymer film dielectric characterization [4]. Combining thermal activation with return voltage to perform stimulated recovery voltage experiments allowed also Japanese researchers to monitor charge injection from needles in polyethylene [60].

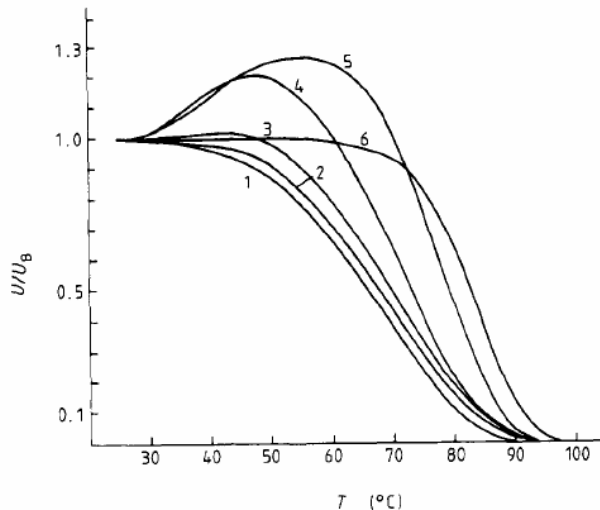


Figure 6. Thermostimulated surface potential on polyacrylonitrile films showing anomalous potential increase [58].

4 CONCLUSIONS AND PROSPECTIVE

The complexity of the response of a material submitted to charging is underlined in this review of potential decay

measurements; it will of course be found in any electrical measurement performed on an insulator, and should be of an interest for the reader, further than the surface potential measurement topic.

Concerning prospective, one can observe that most of the modern techniques for space charge internal mapping in insulating materials rely on current measurements (in closed or open circuit), rather than potential measurements. The domain of potential measurements seems thus to have comparatively narrowed since the seventies, except in the field of surface scanning and 2D mapping, where new microscopy techniques have been developed.

However, potential measurements keep a specific advantage on current measurements wherever static charges are involved: as was mentioned above, they are sensitive to the position of the charges, and not only to their displacement. Combining this particular advantage of potential measurements with a wave technique seems to be an idea that has not been sufficiently exploited, even if some practical difficulties linked to the non-metallization of the upper surface have to be overcome.

For the same reason, a research field seems to be still wide open concerning development of new charge injection transients monitoring techniques, as well new thermally stimulated techniques, using return voltage experiments.

Lastly, an important field in surface potential measurements has recently been opened by Kelvin probe force microscopy, as mentioned in section 1.1.2. Systems that can read and write patterns of charge with a resolution better than 100 nm have been developed, mainly because of their great potential in rewritable digital data storage [61, 62]. An entire treatment of this subject was outside the scope of this article, since it has not for the moment been fully exploited on insulating materials. Such a technique could however give an insight into the microscopic aspects of charge injection and polarization, that might completely change our understanding of these phenomena.

ACKNOWLEDGMENT

We dedicate this work to the memory of Roland Coelho.

REFERENCES

- [1] G. M. Sessler, "Charge distribution and transport in polymers", IEEE Trans. Dielectr. Electr. Insul., Vol. 4, pp. 614-628, 1997.
- [2] D. Marty-Dessus, L. Berquez, A. Petre and J. L. Franceschi, "Space charge cartography by FLIMM: a three-dimensional approach", J. Phys. D: Appl. Phys., Vol. 35, pp. 3249-3256, 2002.

- [3] J. A. Giacometti, S. Fedosov and M. M. Costa, "Corona charging of polymers : recent advances on constant current charging", Brazilian J. Phys., Vol. 29, pp. 269-279, 1999.
- [4] I. Muzikante, E. Fonavs, A. Tokmakov, B. Stiller, L. Brehmer, O. Neilands, and K. Balodis, "Studies of relaxation processes in poled dipolar dye-doped polymeric films", Materials Sci. and Eng. C, Vol. 22, pp. 213-217, 2002.
- [5] E. A. Baum, T. J. Lewis, and R. Toomer, "Further observations on the decay of potential corona charged polyethylene films", J. Phys. D., Vol. 10, pp. 2525-2531, 1977.
- [6] E. A. Baum, T. J. Lewis and R. Toomer, "The lateral motion of charge on thin films of polyethylene terephthalate", J. Phys. D., Vol. 11, pp. 963-977, 1978.
- [7] D. K. Das Gupta, "Decay of electrical charges on organic synthetic polymer surfaces", IEEE Trans. Dielectr. Electr. Insul., Vol. 25, pp. 503-508, 1990.
- [8] H. O. Jacobs, H. F. Knapp, S. Müller and A. Stemmer, "Surface potential mapping : a qualitative material contrast in SPM", Ultramicroscopy No. 69, pp. 39-49, 1997.
- [9] F. Galembeck, C. A. R. Costa, A. Galembeck and M. C. V. M. Silva, "Supramolecular ionics : electric charge partition within polymers and other non-conducting solids", An. Acad. Bras. Cienc., vol. 73, pp. 495-509, 2001.
- [10] P. K. Watson, "The transport and trapping of electrons in polymers", IEEE Trans. Dielectr. Electr. Insul., Vol. 2, pp. 915-924, 1995.
- [11] R. Coelho, B. Aladenize, N. Cornet and J. Bigarré, "On the transient potential in insulators", IEEE Trans. Dielectr. Electr. Insul., Vol. 8, pp. 760-770, 2001.
- [12] I. P. Batra, K. D. Kanazawa and H. Seki, "Discharge characteristics of photoconducting insulators", J. Appl. Phys., Vol. 41, pp. 3416-3422, 1970.
- [13] D. W. Vance, "Surface charging of insulators by ion irradiation", J. Appl. Phys., Vol. 42, pp. 5430-5443, 1971.
- [14] R. H. Young, "Kinetics of xerographic discharge by surface charge injection", J. Appl. Phys., Vol. 72, pp. 2993-3004, 1992.
- [15] G. M. Sessler, C. Alquié and J. Lewiner, "Charge distribution in Teflon FEP (fluoroethylenepropylene) negatively corona-charged to high potentials", J. Appl. Phys., Vol. 71, pp. 2280-2284, 1992.
- [16] H. Amjadi, "The mechanism of voltage decay in corona-charged layers of silicon dioxide during UV irradiation", IEEE Trans. Dielectr. Electr. Insul., Vol. 7, pp. 222-228, 2000.
- [17] G. M. Sessler, "Charge storage in dielectrics", IEEE Trans. Electr. Insul., Vol. 24, pp. 395-402, 1989.
- [18] R. A. Moreno and B. Gross, "Measurement of potential buildup and decay, surface charge density, and charging currents of corona-charged polymer foil electrets", J. Appl. Phys., Vol. 47, pp. 3397-3402, 1976.
- [19] M. Campos and J. A. Giacometti, "Surface-potential decay in insulators with deep traps", J. Appl. Phys., Vol. 52, pp. 4546-4552, 1981.
- [20] G. F. Leal Ferreira and M. T. Figueiredo, "Corona charging of electrets - Models and results", IEEE Trans. Electr. Insul., Vol. 27, pp. 719-738, 1992.
- [21] M. Ieda, G. Sawa and I. Shinohara, "A decay process of surface electric charges across polyethylene film", J. Appl. Phys., Vol. 6, pp. 793-794, 1967.
- [22] M. Ieda, G. Sawa and U. Shinohara, "Decay of electric charges on polymeric films", El. Eng. in Japan, Vol. 88, No. 6, pp. 67-73, 1968.
- [23] T. Mizutani, T. Oomura and M. Ieda, "Surface potential decay in polyethylene", Jap. J. Appl. Phys., Vol. 20, pp. 855-859, 1981.
- [24] R. Coelho, "The electrostatic characterization of insulating materials", J. Electrostatics, Vol. 17, pp. 13-27, 1985.
- [25] R. Coelho, P. Jestin, L. Levy and D. Sarraill, "On the return-voltage buildup in insulating materials", IEEE Trans. Electr. Insul. Vol. 22, pp. 683-690, 1987.
- [26] P. Molinié, M. Goldman and J. Gatellet, "Surface potential decay on corona-charged epoxy samples due to polarization processes", J. Phys. D: Appl. Phys., Vol. 28, pp. 1601-1610, 1995.
- [27] P. Molinié, "Potential decay interpretation on insulating films : necessity of combining charge injection and slow volume polarization processes", Proc. 7th Int Conf. DMMA, Bath, pp. 50-55, 1996.
- [28] P. Llovera and P. Molinié, "Charge injection dynamics on thin films : limited by surface detrapping or by charge drift mobility ? A new contribution using surface potential measurements", 3rd Conf. Société Française d'Electrostatique, Toulouse 2002.
- [29] P. Llovera and P. Molinié, "New methodology for surface potential decay measurements - application to study charge injection dynamics on polypropylene films", IEEE Trans. Dielectr. Electr. Insul., Vol. 11, pp. 1049-1056, 2004.
- [30] H. J. Wintle, "Decay of static electrification by conduction processes in polyethylene", J. Appl. Phys., Vol. 41, pp. 4004-4007, 1970.
- [31] H. J. Wintle, "Surface charge decay in insulators with nonconstant mobility and with deep trapping", J. Appl. Phys., Vol. 43, pp. 2927-2930, 1972.
- [32] T. J. Sonnonstine and M. M. Perlman, "Surface potential decay in insulators with field dependent mobility and injection efficiency", J. Appl. Phys., Vol. 46, pp. 3975-3981, 1975.
- [33] K. J. Kao, S. S. Bamji and M. M. Perlman, "Thermally stimulated discharge current study of surface charge release in polyethylene by corona-generated excited molecules, and the crossover phenomenon", J. Appl. Phys., Vol. 50, pp. 8181-8185, 1979.
- [34] S. Haridoss, M. M. Perlman and C. Carlone, "Vibrationally excited diatomic molecules as charge injectors during corona charging of polymer films", J. Appl. Phys., Vol. 53, No. 9, pp. 6106-6114, 1982.
- [35] R. Coelho and B. Aladenize, "Contribution to the analysis of the decay of charged samples", IEEE CEIDP, pp. 621-625, 1995.
- [36] H. Von Berlepsch, "Interpretation of surface potential kinetics in HDPE by a trapping model", J. Phys. D: Appl. Phys., Vol. 18, pp. 1155-1170, 1985.
- [37] T. K. Saha and Z. T. Yao, "Experience with return voltage measurements for assessing insulation conditions in service-aged transformers", IEEE Trans. Power Del., Vol. 18, pp. 128-135, 2003.
- [38] F. Frutos, M. Acedo, J. C. Filippini, A. Jimenez, J. A. Perez-Gomez and I. Radu, "A preliminary study of the return voltage in polyethylene specimens with the point-plane and plane-plane geometry", Int. Conf. Dielectric Insulation, Budapest, 1997.
- [39] R. Robert and S. M. Berleze, "Integro-differential equation of absorptive capacitors", IEEE Trans. Dielectr. Electr. Insul., Vol. 8, pp. 244-247, 2001.
- [40] D. K. Das Gupta and P. C. N. Scarpa, "Polarization and dielectric behavior of ac-aged polyethylene", IEEE Trans. Dielectr. Electr. Insul., Vol. 3, pp. 366-374, 1996.
- [41] B. Gross, "On discharge voltage and return voltage curves for absorptive capacitors", Phys. Rev., Vol. 62, pp. 383-387, 1942.
- [42] A. K. Jonscher, *Universal relaxation law*, Chelsea Dielectrics Press, London, 1996.
- [43] S. M. Berleze and R. Robert, "Response functions and after effect in dielectrics", IEEE Trans. Dielectr. Electr. Insul., Vol. 10, No. 4, pp. 665-669, 2003.
- [44] A. R. Frederickson, C. E. Benson and J. F. Bockman, "Measurement of charge storage and leakage in polyimides", Nucl. Instrum. and Meth. In Phys. B 208, pp. 454-460, 2003.
- [45] V. Adamec and J. H. Calderwood, "The interpretation of potential decay on the surface of a charged dielectric specimen", J. Phys. D: Appl. Phys., Vol. 20, pp. 803-804, 1987.
- [46] J. A. Malecki, "Linear decay of charge in electrets", Phys. Rev. B, Vol. 59, pp. 9954-9960, 1999.
- [47] A. Crisci, B. Gosse, J. P. Gosse and V. Ollier-Duréault, "Surface potential decay due to surface conduction", Eur. Phys. J. AP., Vol. 4, pp. 107-116, 1998.
- [48] R. M. Hill, "Relaxation of surface charge", J. Phys. C: Solid State Phys., Vol. 8, pp. 2488-2501, 1975.
- [49] F. Mady, R. Renoud and J.-P. Ganachaud, "Modelling of potential decay experiments. Influence of the microscopic charge transport processes on the macroscopic observable", 3rd Conf. Société Française d'Electrostatique, Toulouse 2002.

- [50] S. Sahli, A. Bellel, Z. Ziari, A. Kahlouche and Y. Segui, "Measure and analysis of potential decay in polypropylene films after negative corona charge deposition", J. Electrostatics, Vol. 57, pp.167-181, 2003.
- [51] H. Scher and E. W. Montroll, "Anomalous transit-time dispersion in amorphous solids", Phys. Rev. B, Vol. 12, pp. 2455-2477, 1975.
- [52] V. I. Arkhipov, J. A. Popova and A. I. Rudenko, "Space-charge perturbed dispersive transport in disordered dielectrics", J. Electrostatics, Vol. 18, pp. 23-37, 1986.
- [53] W. Tomaszewicz, "Surface potential decay of disordered solids", J. Electrostatics, Vol. 51-52, pp. 340-344, 2001.
- [54] J. G. Simmons and M. C. Tam, "Theory of isothermal currents and the direct determination of trap parameters in semiconductors and insulators containing arbitrary trap distributions", Phys. Rev. B, Vol. 7, pp. 3706-3713, 1973 .
- [55] H. Von Seggern, "Detection of surface and bulk traps", J. Appl. Phys., Vol. 52, pp. 4086-4089, 1981.
- [56] P. Molinié and P. Llovera, "Mise en oeuvre et interprétation des mesures de potentiel de surface", Proc. 2nd Conf. Société Française d'Electrostatique, Montpellier, pp. 90-95, 2000.
- [57] J. Van Turnhout, *Thermally Stimulated Discharge of Polymer Electrets*, Elsevier, Amsterdam, 1975.
- [58] A. A. Rychkov, V. G. Boitzov and V. V. Shvetz, "Energy distribution of surface traps for elongated in the water and corona charged PTFE films", IEEE 9th Symp. Electrets, pp. 89-92, 1996 .
- [59] H. von Berlepsch, M. Pinnow and W. Stark, "Electrical conduction in acrylonitrile/methylacrylate copolymer films", J. Phys. D: Appl. Phys., Vol. 22, pp. 1143-1152, 1989.
- [60] M. Fukuzawa and M. Iwamoto, "Analysis of the electric charge characteristic in an LDPE film under a needle-plane electrode system by using a TSSP measurement", Electr. Eng. Japan, Vol. 111, 2, pp. 1-10, 1990.
- [61] H. O. Jacobs and G. M. Whitesides, "Submicrometer patterning of charge in thin film electrets", Science, Vol. 291, pp1763-1766, 2001.
- [62] H. O. Jacobs and A. Stemmer, "Measuring and modifying the electric surface potential distribution on a nanometre scale : a powerful tool in science and technology", Surf. and Interf. Analysis, Vol. 27, pp361-367, 1999.



Philippe Molinié was born in 1965 at Suresnes, France. He received in 1987 his engineering degree from the Ecole Supérieure d'Electricité (Supélec). He obtained the Ph.D. degree from the Université Pierre et Marie Curie (Paris VI University) in 1992 on the topic of neutralization of the electrical charge deposited by corona discharge on an insulating surface. He is teacher-cum-researcher since 1995 at the Ecole Supérieure d'Electricité. His technical fields of interest are surface potential measurements, dielectric materials characterization, charge injection into insulating materials and electrostatics.